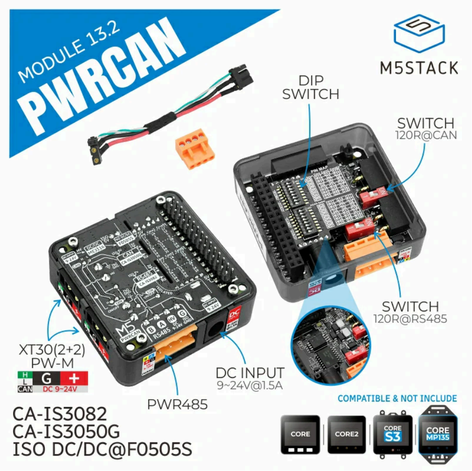
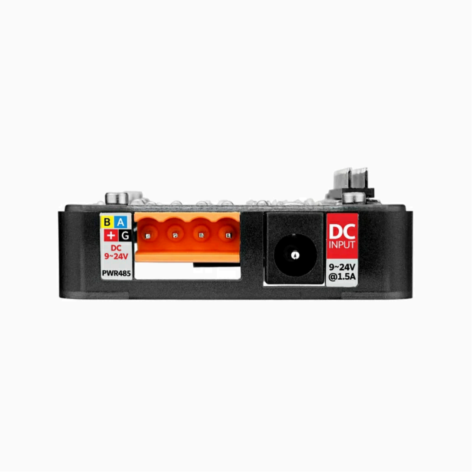
Module13.2 PwrCAN

SKU:M139


MODULE 13.2
PWRCAN
M5STACK
DIP SWITCH
SWITCH 120R@CAN
SWITCH 120R@RS485
XT30(2+2) PW-M
H L G +
CAN DC 9~24V
DC INPUT 9~24V@1.5A
PWR485
CA-IS3082
CA-IS3050G
ISO DC/DC@F0505S
COMPATIBLE & NOT INCLUDE
CORE
CORE2
CORE S3
CORE MP135


B A
+ G
DC 9~24V
PWR485
DC INPUT
9~24V @1.5A

PWRCAN
MODULE 13.2

Description

Module13.2 PwrCAN is a versatile module designed for the PwrCAN bus, which integrates isolated CAN communication and a DC 9 ~ 24V power supply bus. The module also includes Pwr485 (isolated) bus functionality and provides isolated 5V power supply to the M5 host. The CAN communication part uses the CA-IS3050G isolated transceiver, and the RS485 part uses the CA-IS3082W isolated transceiver. The GPIOs associated with CAN and RS485 communication can be selected via DIP switches, and 120-ohm terminal resistors can be connected in parallel at the CAN and RS485 outlets via DIP switches. The module's power bus supports a wide input voltage range of DC 9 ~ 24V, with the DC jack directly connected to the HT3.96 and XT30 power sections. The module provides power to the M5 host through the built-in isolated power module F0505S-2WR3. This module is suitable for applications such as robot control, protocol conversion, industrial automation, vehicle communication systems, intelligent transportation, and building automation.

Note

Note

The power section of the XT30 and HT3.96 interfaces is directly connected to the DC input with no voltage step-down — the output voltage equals the input voltage. Before connecting, verify that the device supports the operating voltage to prevent damage caused by overvoltage.

Features

- Isolated CAN bus (dual-channel)
- Isolated RS485 bus
- Signal isolation
- Isolated power supply
- Multiple power interfaces
- Wide input voltage range (9-24V)
- CAN/RS485 communication pin switching DIP switch

Includes

- 1 x Module13.2 PwrCAN
- 1 x HT3.96-4P
- 1 x XT30(2+2)PW-M cable

Applications

- Robot control
- Protocol conversion
- Industrial automation
- Vehicle communication systems
- Intelligent transportation
- Building automation

Specifications

Specification	Parameters
Supported Protocols	CAN & RS485 bus communication protocols
CAN Communication	CA-IS3050G@dual-channel
RS485 Communication	CA-IS3082W@single-channel
CAN Interface	XT30(2+2)PW-M horizontal solder board male header
RS485 Interface	HT3.96-4P
CAN Bus Speed	Up to 1Mbps
RS485 Bus Speed	Up to 500Kbps
CAN Supported Nodes	110
RS485 Supported Nodes	256
Voltage Input Range	9-24V
DC Power Port	Input: DC 9–24V, compatible with DC5521 (5.5 × 2.1 mm) plug, center positive
Power Supply Methods	DC power jack (DC5521 5.5 x 2.1mm, inner positive outer negative), 485 interface (HT3.96) power supply, CAN interface (XT30) power supply
Operating Temperature	0-40°C
Product Size	54.0 x 54.0 x 19.7mm
Product Weight	27.8g
Package Size	132.0 x 95.0 x 21.0mm
Gross Weight	51.2g

RS485 Communication Test

Communication Distance	Data Rate
30 meters	Maximum data rate 512Kbps, normal transmission and reception, 0% packet loss, 0% error rate
50 meters	Maximum data rate 512Kbps, normal transmission and reception, 0% packet loss, 0% error rate
100 meters	Maximum data rate 512Kbps, normal transmission and reception, 0% packet loss, 0% error rate

CAN Communication Test

Communication Distance	Data Rate
10 meters	Data rate 1000Kbps, normal transmission and reception, 0% packet loss, 0% error rate
30 meters	Data rate 500Kbps, normal transmission and reception, 0% packet loss, 0% error rate
50 meters	Data rate 500Kbps, normal transmission and reception, 0% packet loss, 0% error rate
100 meters	Data rate 250Kbps, normal transmission and reception, 0% packet loss, 0% error rate

Load Capacity Test

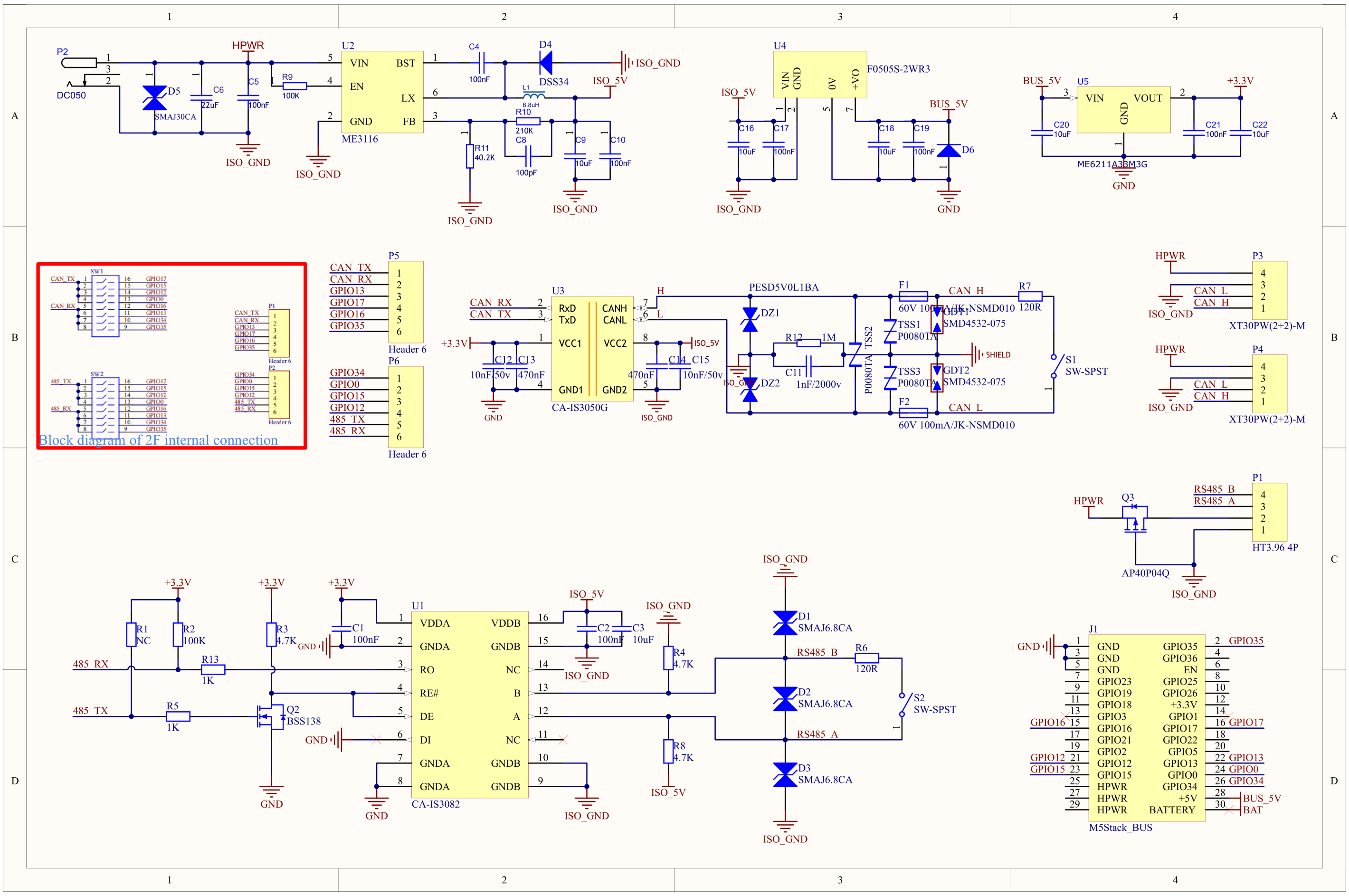
Parameter Category	Specifications
Output Load Capacity	Isolated output maximum current: DC: 4.70V@218mA Non-isolated output maximum current: DC: 4.70V@1.1A

Power Consumption Test

Parameter Category	Specifications
Standby Current	DC 9V@30.94mA DC 12V@23.57mA DC 24V@10mA
Operating Current (with host)	DC 9V@78.60mA DC 12V@65.54mA DC 24V@31mA

Schematics

- Module13.2 PwrCAN Schematics PDF


P2
DC050
HPWR
D5
SMAJ30CA
C6
22uF
C5
100nF
R9
100K
ISO_GND
U2
VIN
EN
GND
BST
LX
FB
ME3116
C4
100nF
D4
DSS34
ISO_5V
L1
6.8uH
R10
210K
C8
100pF
R11
40.2K
C9
10uF
C10
100nF
U4
F0505S-2WR3
VIN
GND
0V
+VO
C16
10uF
C17
100nF
C18
10uF
C19
100nF
D6
BUS_5V
GND
U5
VOUT
C20
10uF
C21
100nF
C22
10uF
ME6211A33M3G
+3.3V
Block diagram of 2F internal connection
SW1
SW2
CAN_TX
CAN_RX
GPIO13
GPIO17
GPIO16
GPIO35
Header 6
P5
P6
GPIO34
GPIO0
GPIO15
GPIO12
485_TX
485_RX
U3
RxD
TxD
VCC1
GND1
CANH
CANL
VCC2
GND2
CA-IS3050G
C12
10nF/50v
C13
470nF
C14
470nF
C15
10nF/50v
PESD5V0L1BA
DZ1
DZ2
R12
1M
C11
1nF/2000v
TSS1
TSS2
TSS3
P0080TA
F1
F2
60V 100mA/JK-NSMD010
GDT1
GDT2
SMD4532-075
SHIELD
CAN_H
CAN_L
R7
120R
S1
SW-SPST
P3
P4
XT30PW(2+2)-M
P1
RS485_B
RS485_A
Q3
AP40P04Q
HT3.96 4P
R1
NC
R2
100K
R3
4.7K
R13
1K
R5
1K
Q2
BSS138
485_RX
485_TX
U1
VDDA
GNDA
RO
RE#
DE
DI
VDDB
GNDB
NC
B
A
CA-IS3082
C1
100nF
C2
100nF
C3
10uF
R4
4.7K
R8
4.7K
D1
D2
D3
SMAJ6.8CA
R6
120R
S2
SW-SPST
J1
M5Stack_BUS
GPIO23
GPIO19
GPIO18
GPIO3
GPIO21
GPIO2
HPWR
BATTERY
GPIO36
GPIO25
GPIO26
GPIO22
GPIO1
GPIO5
+5V
BAT

PinMap

M5-Bus

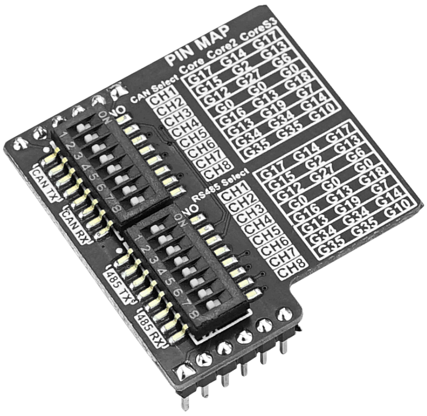
Switch

> The pins marked `SW` on the M5-Bus below can be switched via DIP switches to adapt to different main controllers.

PIN	LEFT	RIGHT	PIN
GND	1	2	CAN_RX/485_RX (SW)
GND	3	4	
GND	5	6	
	7	8	
	9	10	
	11	12	3V3
	13	14	
CAN_RX/485_RX (SW)	15	16	CAN_TX/485_TX (SW)
	17	18	
	19	20	
CAN_TX/485_TX (SW)	21	22	CAN_RX/485_RX (SW)
CAN_TX/485_TX (SW)	23	24	CAN_TX/485_TX (SW)
	25	26	CAN_RX/485_RX (SW)
	27	28	5V
	29	30	BAT

Note

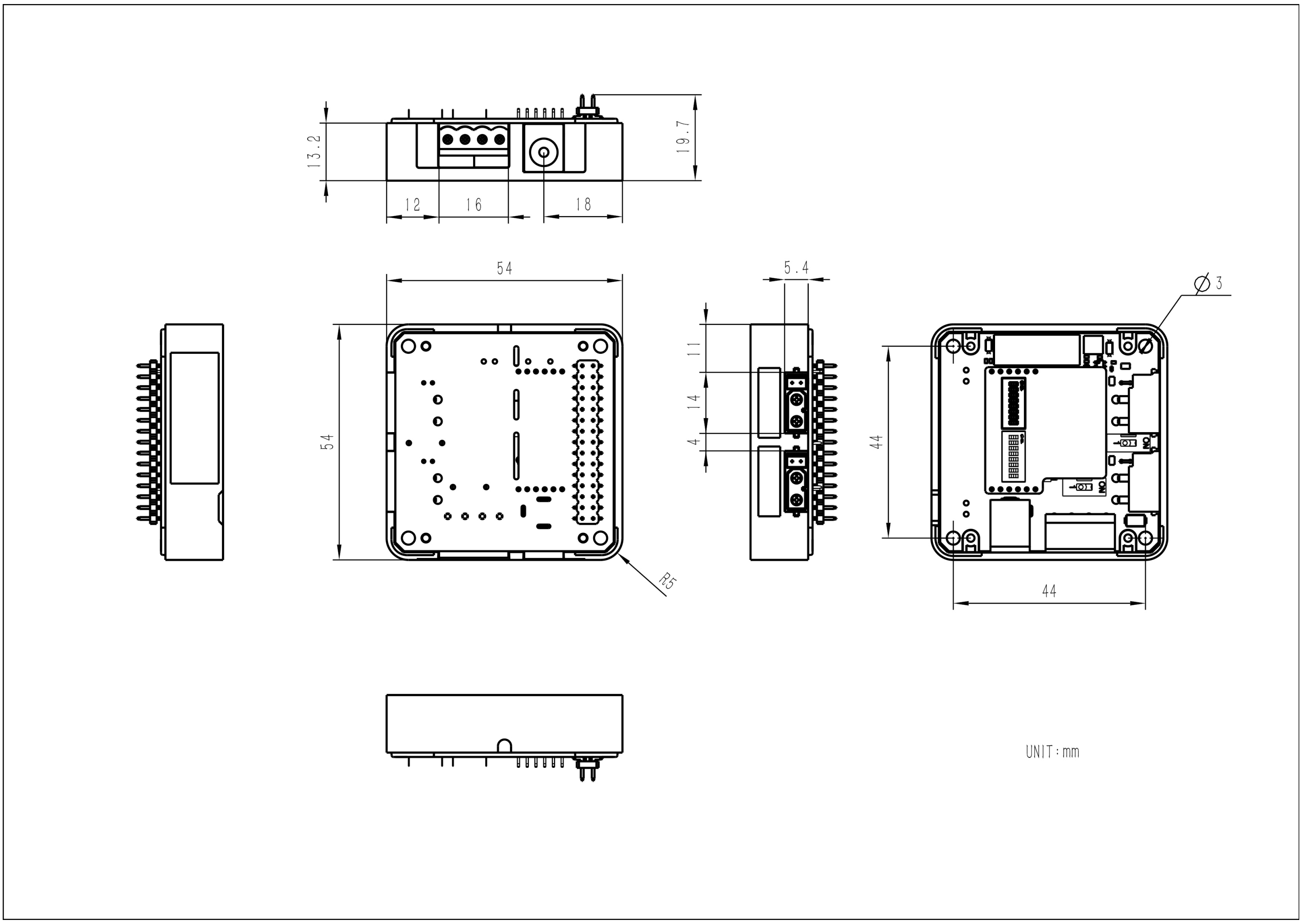
The onboard DIP switch can be used to change the I/O used by the CAN/R485 bus. Switching to the ON side indicates the line is connected. Please note when using the DIP switch to change I/O, the I/O pins used by the CAN bus and the RS485 bus cannot be duplicated, otherwise it will cause conflicts and lead to communication abnormalities.



CAN Select	Core	Core2	CoreS3
CH1	G17	G14	G17
CH2	G15	G2	G13
CH3	G12	G27	G6
CH4	G0	G0	G0
CH5	G16	G13	G18
CH6	G13	G19	G7
CH7	G34	G34	G14
CH8	G35	G35	G10

RS485 Select			
CH1	G17	G14	G17
CH2	G15	G2	G13
CH3	G12	G27	G6
CH4	G0	G0	G0
CH5	G16	G13	G18
CH6	G13	G19	G7
CH7	G34	G34	G14
CH8	G35	G35	G10

Model Size


13.2
19.7
12
16
18
54
54
5.4
11
14
4
44
44
Ø3
R5
UNIT：mm

Datasheets

- CA-IS3050G(CAN)
- CA-IS3082W(RS485)

Softwares

Arduino

- Module13.2 PwrCAN Transceiver Test
- Module13.2 PwrCAN RS485 Example

UiFlow2

- Module13.2 PwrCAN UiFlow2 Docs

Video

pwrcan视频.mp4